

Task 5A:

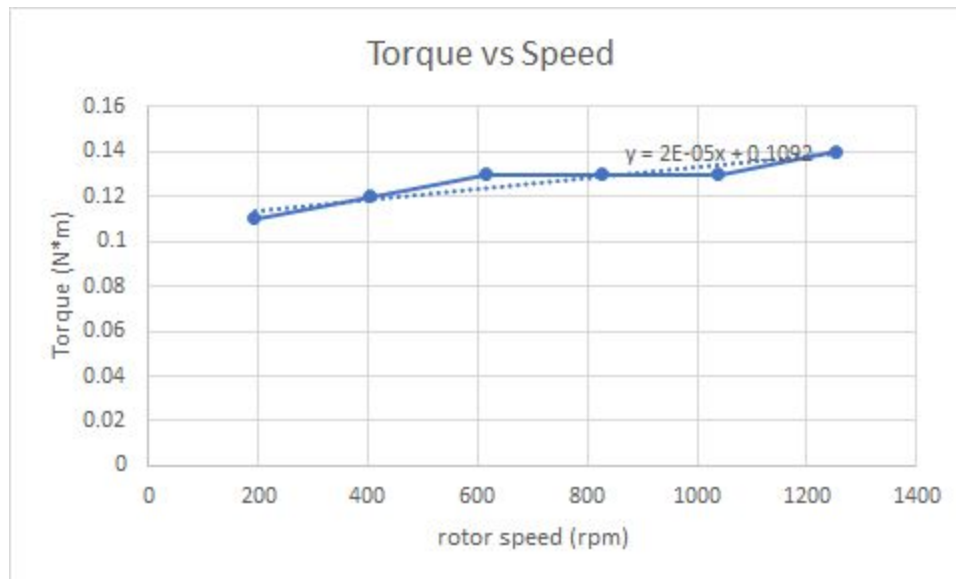
$$(1) \quad R_a = (0.85 + 0.785 + 0.733)/(3) = 0.79\Omega$$

$$(2) \quad K_t = (V_a - I_a R_a)/(\omega r) = 0.024$$

$$K_v = 0.023$$

It can be seen clearly that these two constants are nearly identical in value. This is as expected because the constant K_v is the same representation when the load is disconnected from the circuit.

(3)



The loss that we can see in the curve of torque vs speed can mostly be attributed to sliding friction of the motor with the walls of the machine

$$(4) \quad (J_{rotor})(d\omega r/dt) = (J_{rotor}/K_t)(dE_a/dt) = T_{fric}(\omega r)$$

Using this relationship and the graph from the previous section, we can estimate the moment of inertia of the system.

$$J_{system} = 0.956$$

$$(5) \quad L_a = (\Delta t)(V_a - R_a I_a - K_t - \omega r)/(I_2 - I_1)$$

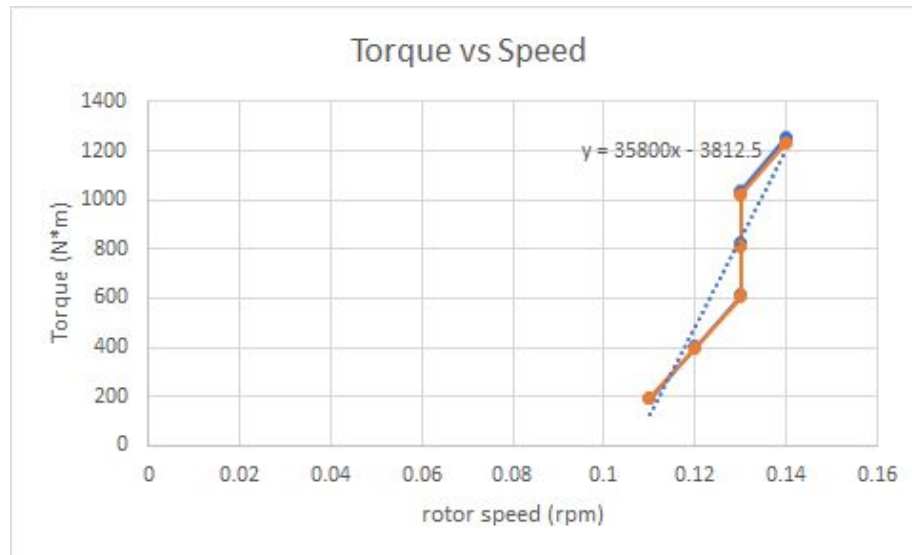
Using this equation from the pre-laboratory section and the recorded data from task 3, we can estimate the armature inductance of the circuit. We must also take into account the ripple voltage information from Task3A.

$$L_a = 30.3H$$

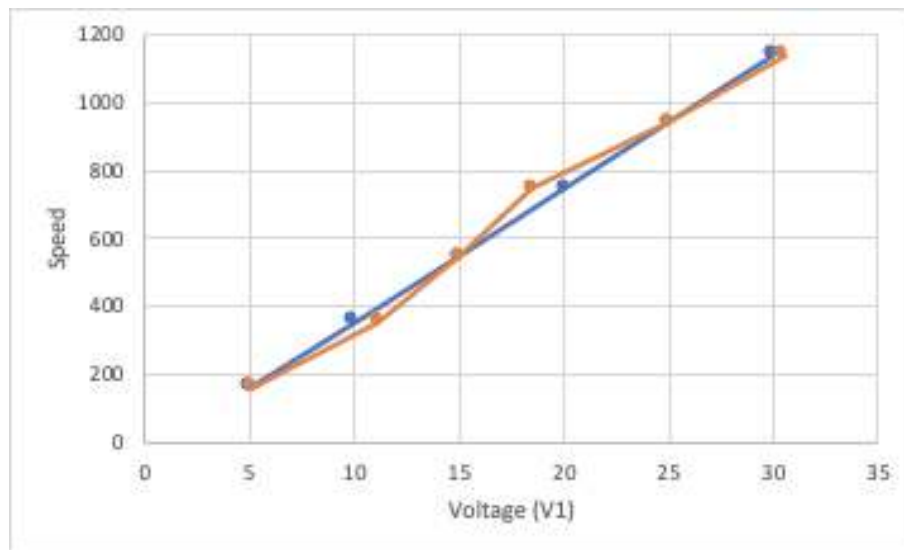
(6) See page 18 of attached data and figures

Task 5B:

- (1) We assumed that we are using the no-load test plot for comparison. The comparison between calculated and measured plot is very similar, because K_t and R_a was calculated with the same set of data.

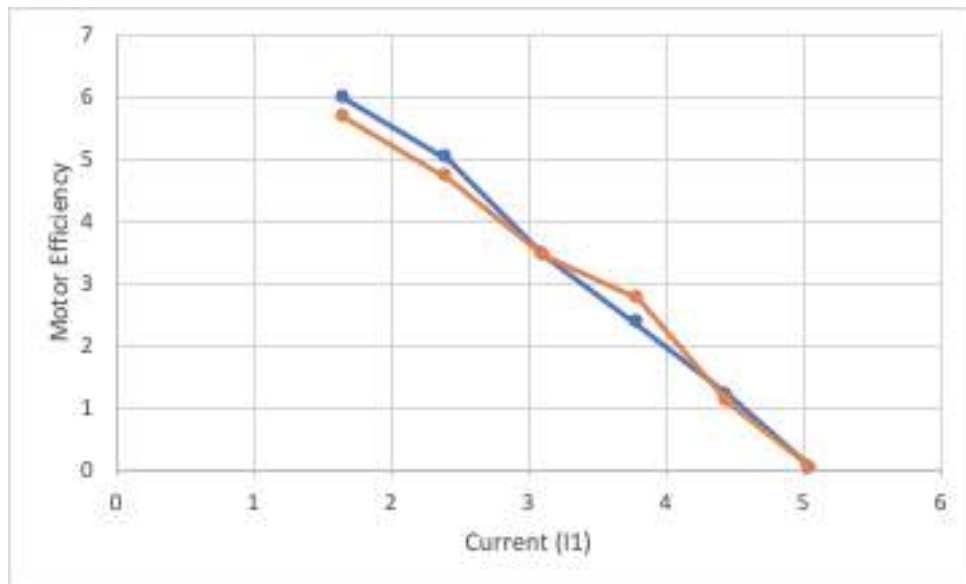


(2)



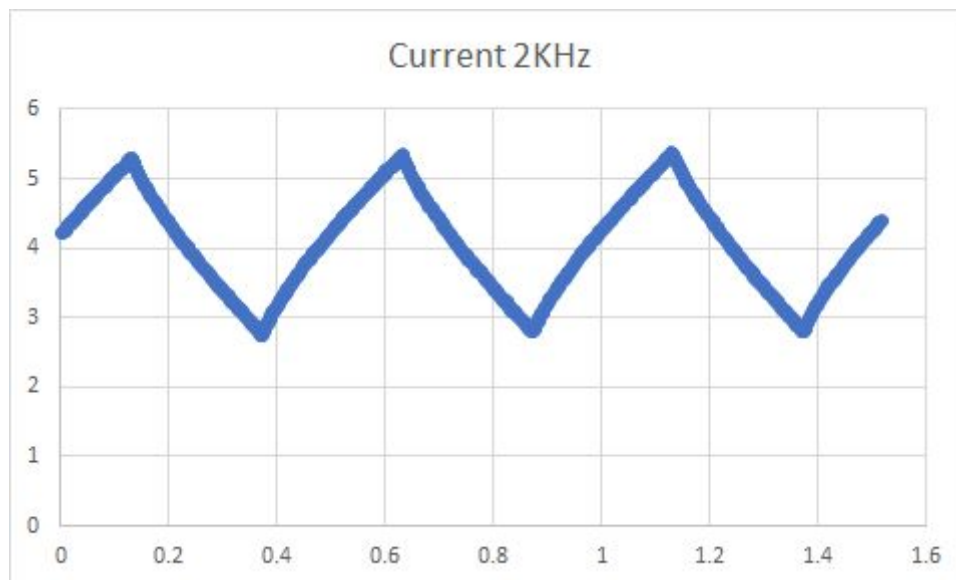
We can see from this plot that the relationship between speed and voltage is linear. As the applied voltage is increased over evenly spaced intervals, the corresponding speed increases linearly to match this trend. The trendline of the equivalent circuit tracks our measured results fairly closely. However, you can still see distinct differences in the plotted data.

(3)

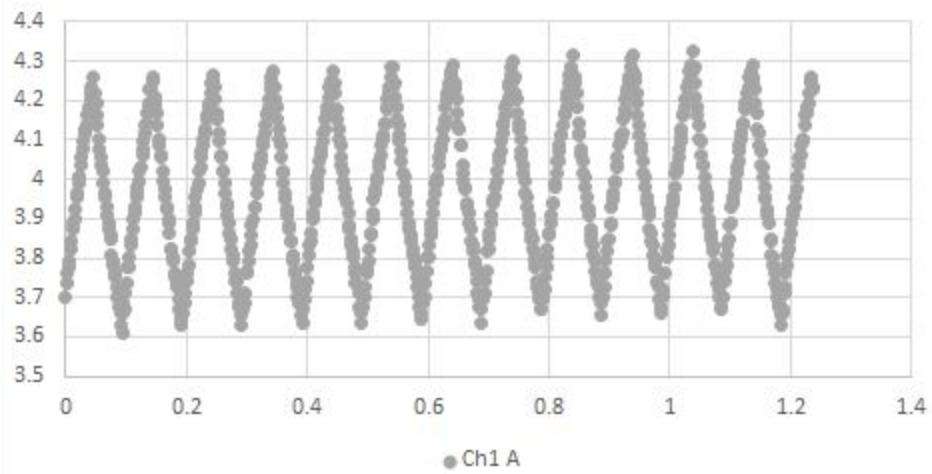


Again we can see from this plot that the equivalent circuit is a good approximation for the measured data. The motor efficiency decreases as current increases. We can see from this plot that the motor has maximum efficiency when the current is lowest, in this case at 1.67A.

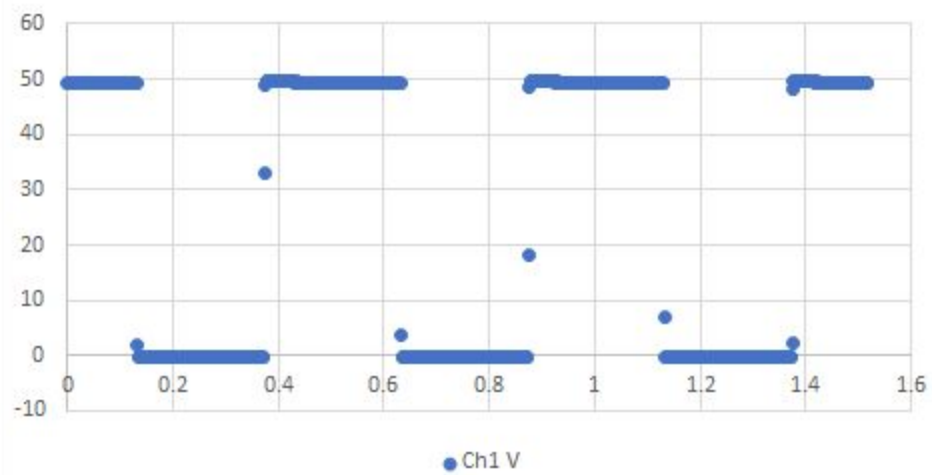
(4) At higher frequencies, the voltage drop across the inductor will be bigger. As a result, the voltage across the motor will be smaller, and the armature current will also be smaller.

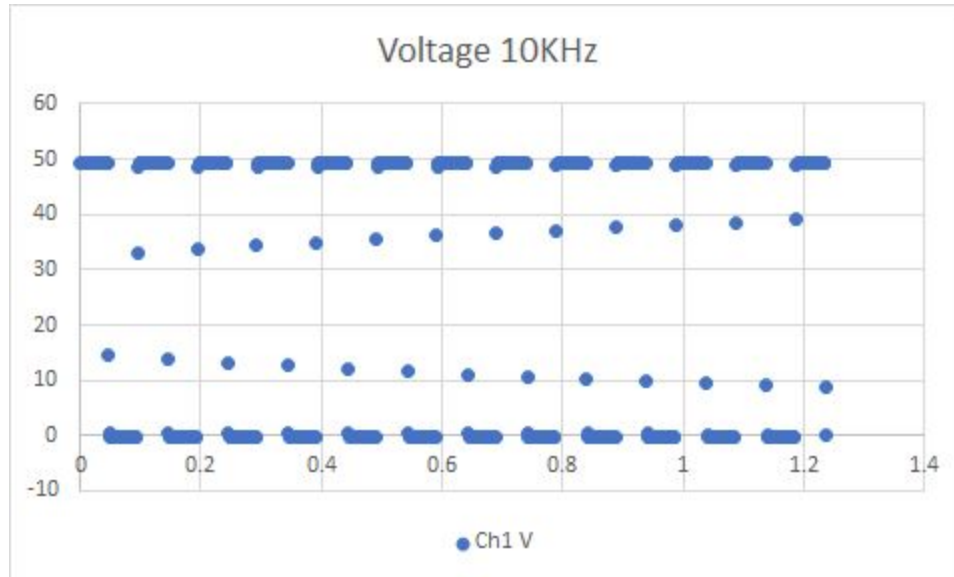


Current 10KHz



Voltage 2KHz





Task 5D:

(1) It can be seen from all plots in the previous section that the equivalent circuit model was a good estimate for the actual results obtained from measurement. This gives us confidence in the techniques we used from ELEC 342 to obtain the values for the various circuit components in the equivalent model.

(2) In our opinion, the method of speed control used in Task 3/4 is a more practical method of examining large motors and/or robotic machines. We feel that the method using the PWM and universal motor controller box is superior because it allows for very precise and highly accurate measurements to be made at various levels of analysis.

(3) See page 19 of attached data and figures

Summary:

This lab served as a good foundation for our introduction to DC motors and rotating machines. We learned different methods and approaches for how to analyze the power transmission and losses within the circuit. We applied techniques and tools learned during ELEC 342 lectures to determine the values of different components for our equivalent circuit model. We saw how this model compared to the actual measured results. The model tracked the measurements very closely however there were some discrepancies which is as expected because this model is just an approximation of the actual circuit.